

Leadership Initiatives AI & Emerging Technology Internship

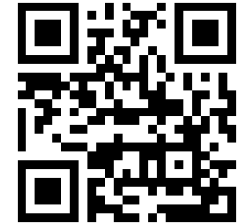
Business Process (Analysis & BPA) Improvement (BPI)

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Slides: <https://tinyurl.com/EspinosaLeadBPI>



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Prof. Espinosa holds a [Ph.D.](#) and [Master of Science in Information Systems](#) from the [Tepper School of Business](#) at [Carnegie Mellon University](#), a [Master of Business Administration](#) from [Texas Tech University](#), and a [Mechanical Engineering](#) degree from [Pontificia Universidad Catolica del Peru](#). He is the architect and implementor of [Kogod's MS Analytics Program](#) and the undergraduate specialization in [Business Analytics](#). He is also the curriculum architect for the [IT&A programs](#) and specializations. He has over 70 published journal articles, conference proceedings, books, and book chapters. He has co-authored two research books, one on work coordination across time zones and a 2-volume book on big data and analytics for service delivery. He has also written a book titled [“Predictive Analytics and Machine Learning for Managers”](#) with practical applications in R. His research focuses on coordination and performance in technical projects across global boundaries, mainly distance and time zones. More recently, he has developed social network and graph analytics methods to represent team knowledge quantitatively and visually. He has 70 peer-reviewed articles, many of them published in leading scholarly journals, including [Management Science](#), [Organization Science](#), [Information Systems Research](#), the [Journal of Management Information Systems](#), [IEEE Transactions on Software Engineering](#), [IEEE Transactions on Engineering Management](#), [Communications of the ACM](#), [Human Factors, Information, Technology and People](#), and [Software Process: Improvement and Practice](#). His work has also been featured in leading academic conference proceedings. He teaches predictive analytics with machine learning, social & organizational network analytics, R programming, information technology foundations, business process analysis, and programming for business applications. He has received multiple service, research, and teaching awards, including [“American University 2025 Outstanding Teacher”](#), [“2022 Kogod Graduate Professor of the Year”](#), and [“American University 2022 Green Teacher”](#). He has several years of work experience, first as a mechanical design engineer and later as a senior manager, VP, and CFO with international organizations, directly supporting, supervising, and formulating policy for finance and global IT, data management, and analysis, where he designed and developed many software applications to support geographically distributed work.

Hungry anyone?

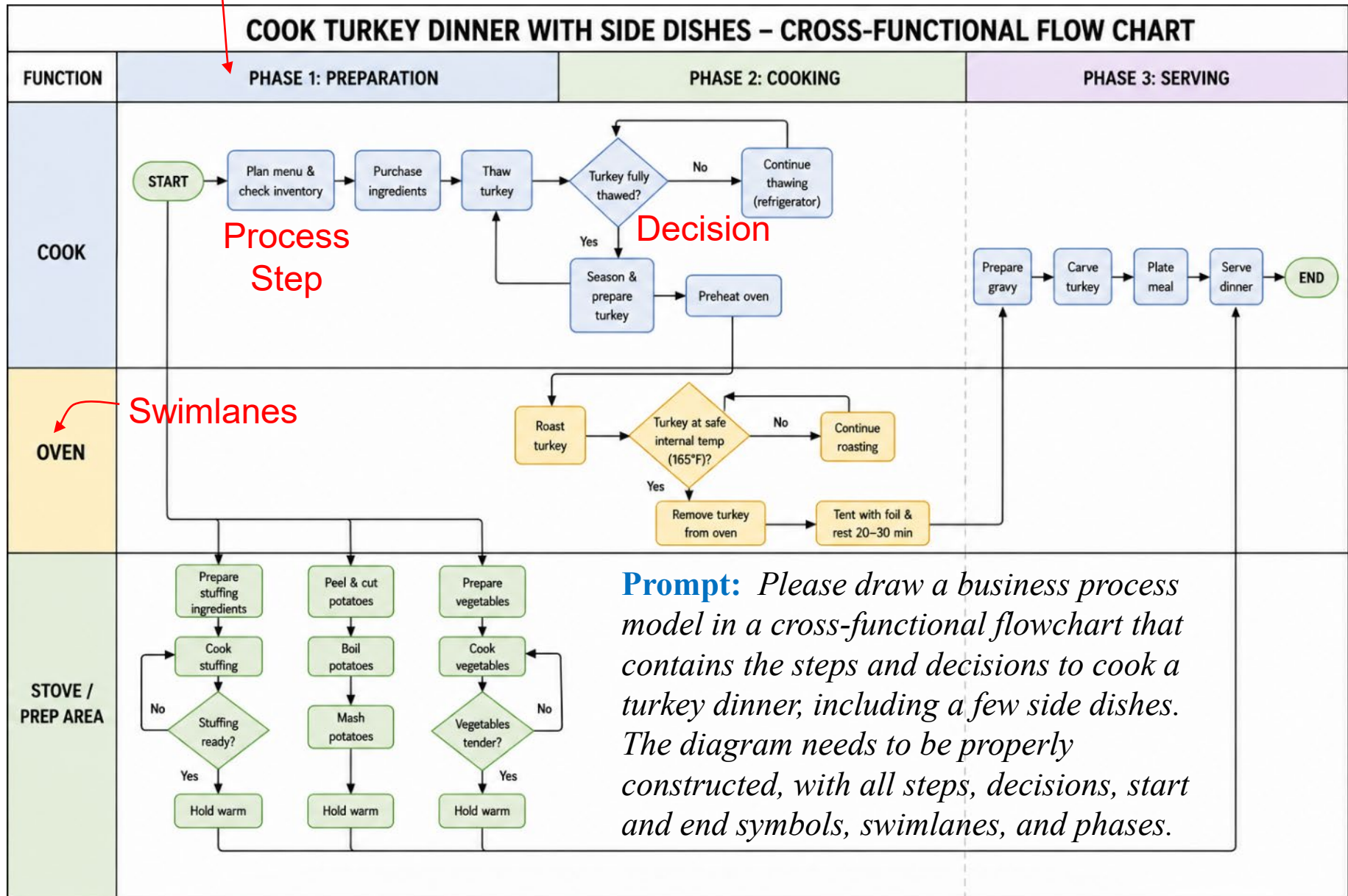


Describe the process to prepare a turkey dinner
- i.e., all the necessary steps, decisions,
and the proper sequence (i.e., workflow).

ChatGPT Solution

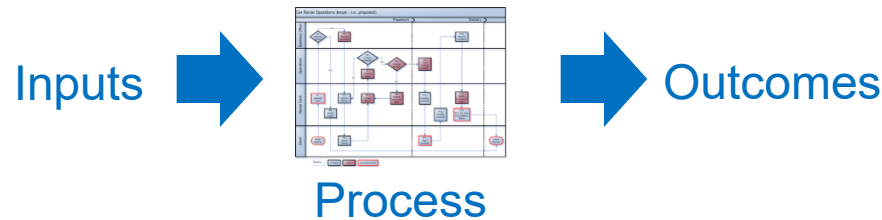
Phases

COOK TURKEY DINNER WITH SIDE DISHES – CROSS-FUNCTIONAL FLOW CHART



What is a **Process**?

- **System** = *Input → Processes → Outcome*
- **Process** = *the steps, decisions, and workflows to convert inputs into outcomes.*



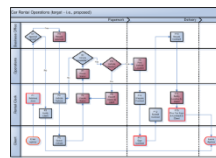
- There are many **types** of systems and processes: e.g., an internal computer, biological, combustion engine, cooking dinner, voting election, business, etc.
- A **Business Process (BP)** is “a set of logically related tasks performed to achieve a defined business outcome.”
(Tom Davenport) [Big Bang Theory](#)

Business Process Examples

Business Domain	Client Facing	Internal
Sales	Point of sales, returns	Inventory management
Online orders	Shopping, checkout	Fulfillment, deliveries
Air travel	Reservations, check-in	Baggage handling
Hotel management	Check-in, check-out	Housekeeping services
Restaurant	Reservations, food orders	Food preparation
Banking	ATM transactions, instrument sales (e.g., CDs)	Account management, cash management
Loans	Applications, payments	Evaluation, underwriting
Insurance	Policy purchases, claims	Underwriting, claim processing

Business Process Analysis (BPA) & Business Process Improvement (BPI)

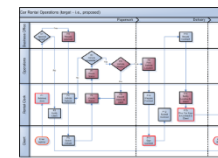
- *Business Process Analysis (BPA) is the study of the process steps, decisions, and workflows involved in a current (i.e., **baseline**) business process,*
- *Undertaken to document and understand it*
- *And then conceptualize and formulate a new or revised (i.e., **target**) business process → resulting in Business Process Improvements (BPI).*



Baseline
Process



Analysis



Target
(Improved)
Process

BPA Objectives

- Identify and understand a client's **business problem**
- Elicit, discover, gather & document **client or product requirements**
- Analyze/understand the **baseline BP** (i.e., the present work)
- Conceptualize business process **improvements and solutions**
- Formulate a **target BP**

Also, often:

1. If some improvements will be handled by a **system** (or **application**), specify its **functionality** to support the target BP
2. Identify and specify the basic **data requirements** to support the target BP and system functionality
3. **Document** the target BP with descriptions, diagrams, and models → important to achieve common ground with your client

Drivers of Business Process Improvement

- Process improvements, extensions or enhancements do **not** happen in a **vacuum**.
- What **drives** a process improvement project?
 - Need to get the perspective of anyone affected (i.e., **stakeholders**)
 - The **business case** & **problem** → provide the focus of attention
 - Process improvement **goals** → general improvements are OK, but they are most effective when they are targeted to specific **process improvement goals**:
 - **Outcome goals** → what we **get** from the process (e.g., increased output, improved products/service quality, etc.)
 - **Process goals** → **how** we achieve the outcomes (e.g., efficiency, speed, low cost, better HR utilization, etc.)

Process Improvement Goals

- Improvements must be driven by specific **process improvement goals** discussed with your client.
- Which can of many types, including:

Outcomes:

- ✓ Increased **revenues**
- ✓ Increased **profits**
- ✓ Reduced operational **costs**
- ✓ Improved client service or **product quality**

Process:

- ✓ More effective **human resource utilization**
- ✓ Increased **efficiencies**
- ✓ Improved **responsiveness** and **agility** to change

Tom Davenport's: 5 Steps to Business Process Improvement

BPI is driven by a **business vision** driven by **specific objectives** (e.g., cost reduction, quality improvement, etc.)

It's important to establish and measure the baseline (**as-is**) process so that the extent of the **improvement** can be **compared** and **quantified**.

Improvement is not a one shot deal, but is **iterative** and **evolutionary**. Focus initially on low hanging fruit, **high impact** areas, customer satisfaction

Develop the business **vision** and process **objectives**



Identify the processes to be redesigned



Understand **baseline** processes



Identify **IT** levers



Design and **develop target** (i.e., improved) process

High Impact Approach: focus on most important parts of the process with potential to reach the vision;
Exhaustive Approach: identify all areas of process improvement and prioritize.

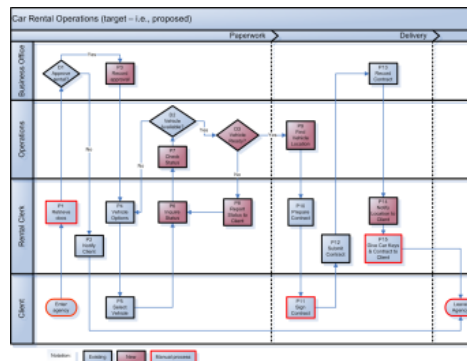
Awareness of IT capabilities are key in improving implementation **speed**, **effectiveness** and **efficiency**. Focus on the **information content** of the process.

Business Process Modeling



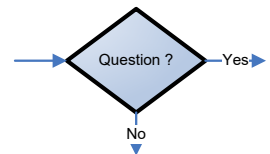
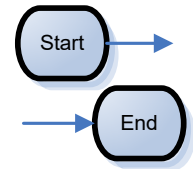
A **BP model (BPM)** is a diagram with a visual representation of the business model

- A **BPM** is the vision of “**the work**” – i.e., how it is done (baseline) and how it will be done (target).
- It **fosters communication** among the consulting team members and with the client – a picture is worth a 1000 words.
- It is necessary to **validate the baseline** and **describe/document** target BP solutions with the client.



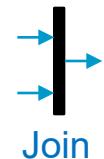
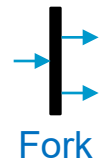
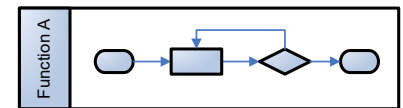
Cross-Functional BPM Main Symbols

- **Terminator:** all processes must have **ONE Start** and **ONE End** point/symbols. **Start** symbols have **NO input** and exactly **1 output**; **End** symbols have **1 or more inputs** and **no output**.
- **Process step:** a specific step in the process – it can have **1 or more inputs**, but it can **ONLY** have **1 output** (if more than 1 output is needed, use a decision or fork symbol). **Steps cannot stand alone, i.e., there cannot be a step without at least 1 input and without 1 output.**
- **Pre-defined process step:** a group of steps and flows grouped into a sub-process. They are useful to simplify the diagramming of complex processes.
- **Decision:** a question or a branching in the process flow it **CAN** have 1 or more inputs and **MUST** have **1 labeled output** for **EACH** possible decision outcome. **There cannot be a decision without at least 1 input and without at least 1 output.**



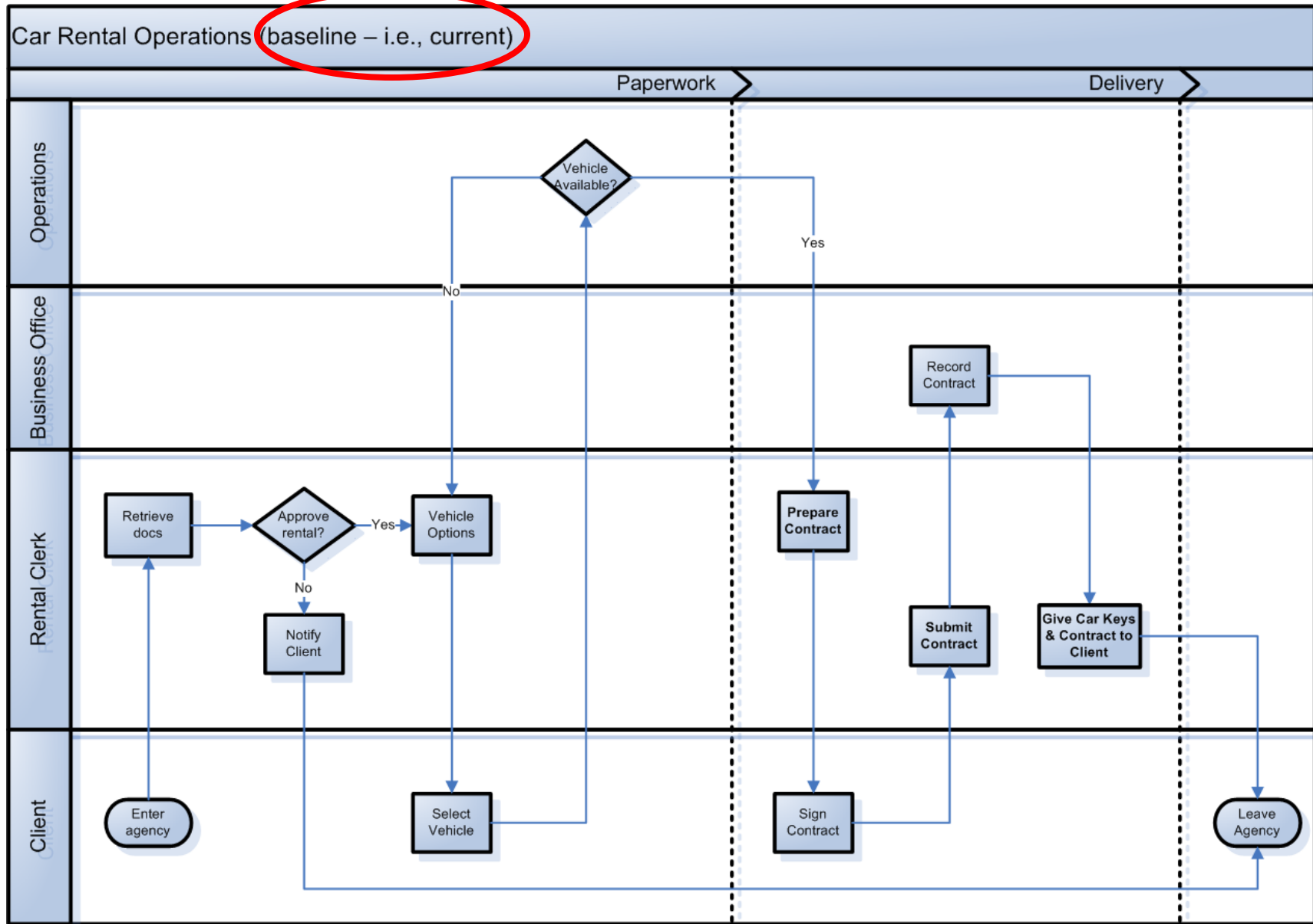
Cross-Functional BPM Main Symbols (cont'd.)

- **Process flow:** a directed arrow that specifies the process sequence. A flow MUST come from a symbol and connect to another symbol, i.e., flows CANNOT be disconnected.
- **Functional bands or swim lanes:** show which departments, units or functional roles are associated with different parts of the process
- **Phase separators:** differentiate different phases of the process sequence (i.e., timing)
- **Parallel Processing:** use these to branch out into multiple parallel flows or to merge them into a single flow.
 - A **Fork** splits a flow into 1 or more parallel flows. Needed when the flow splits without a decision to process different paths in parallel.
 - A **Join** merges one or more parallel flows into 1

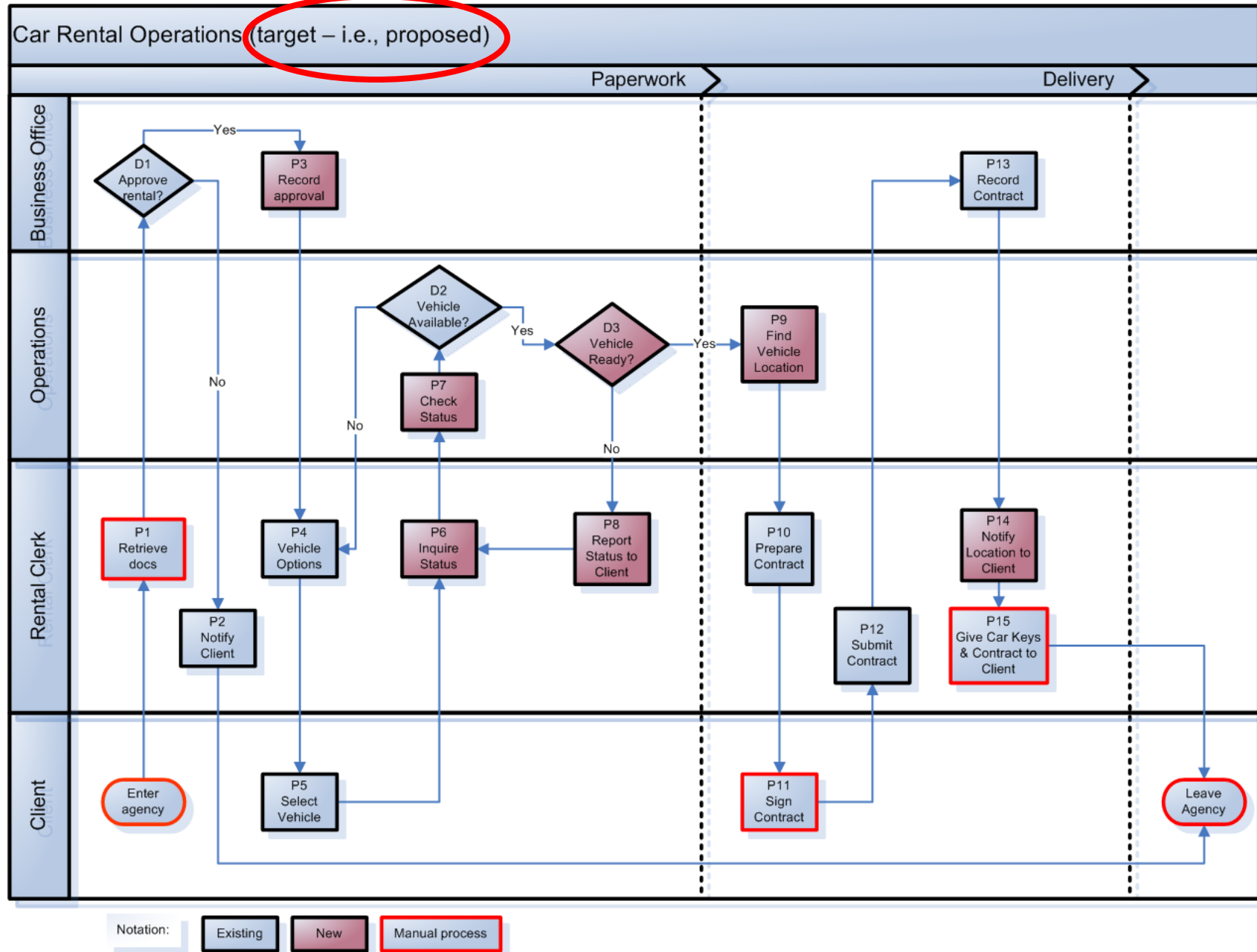


BPM Example - Current

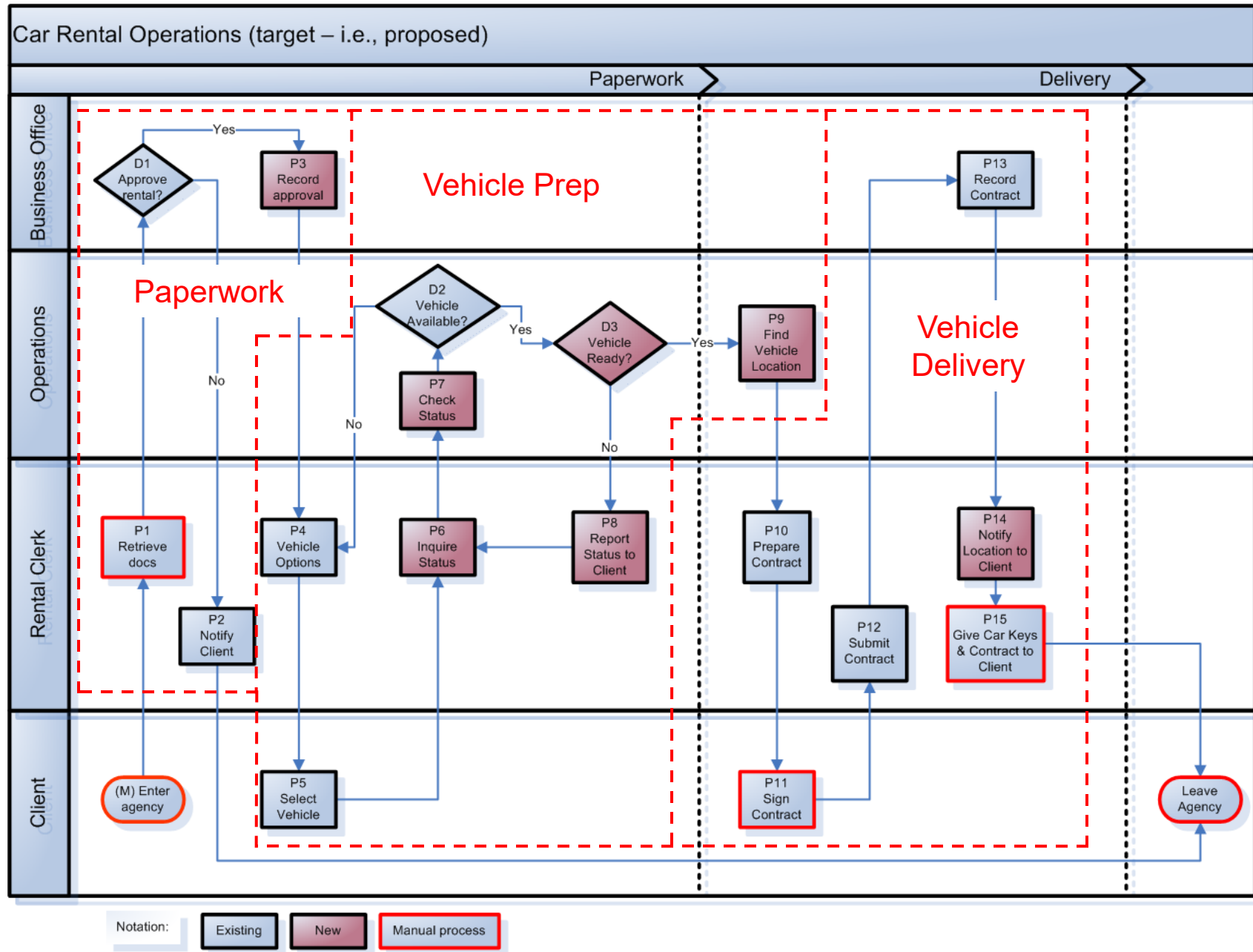
Cross-Functional Flowchart



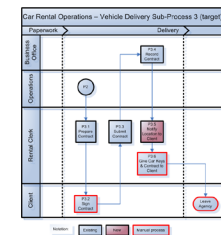
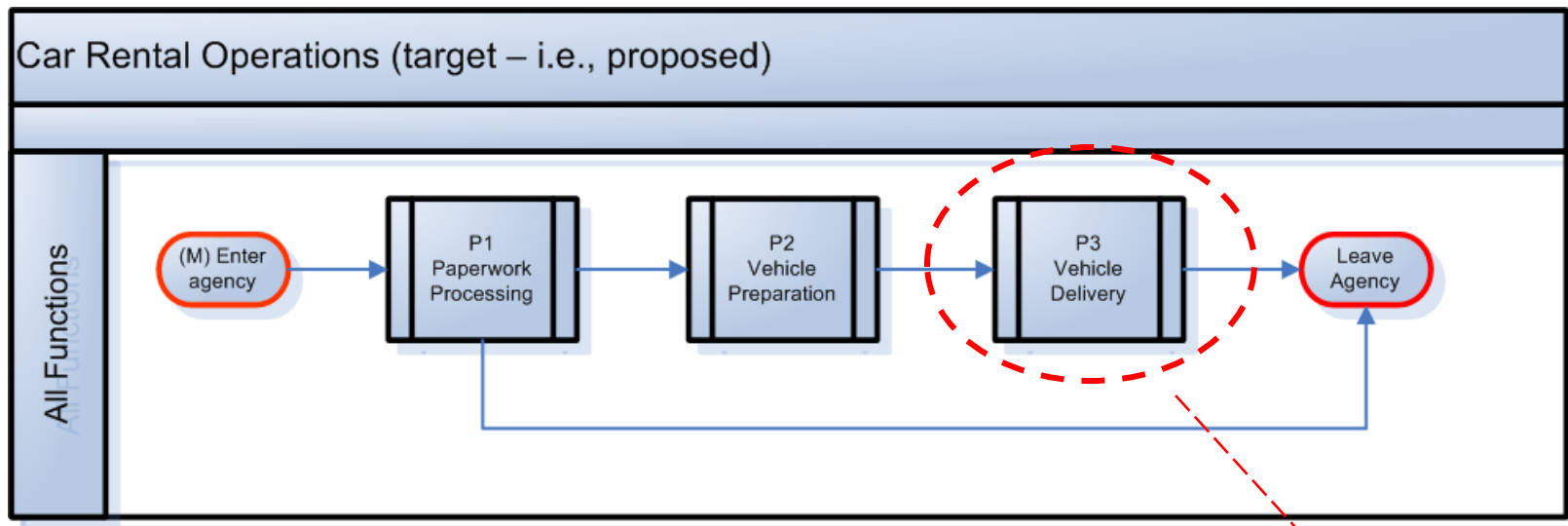
BPM Example – Proposed



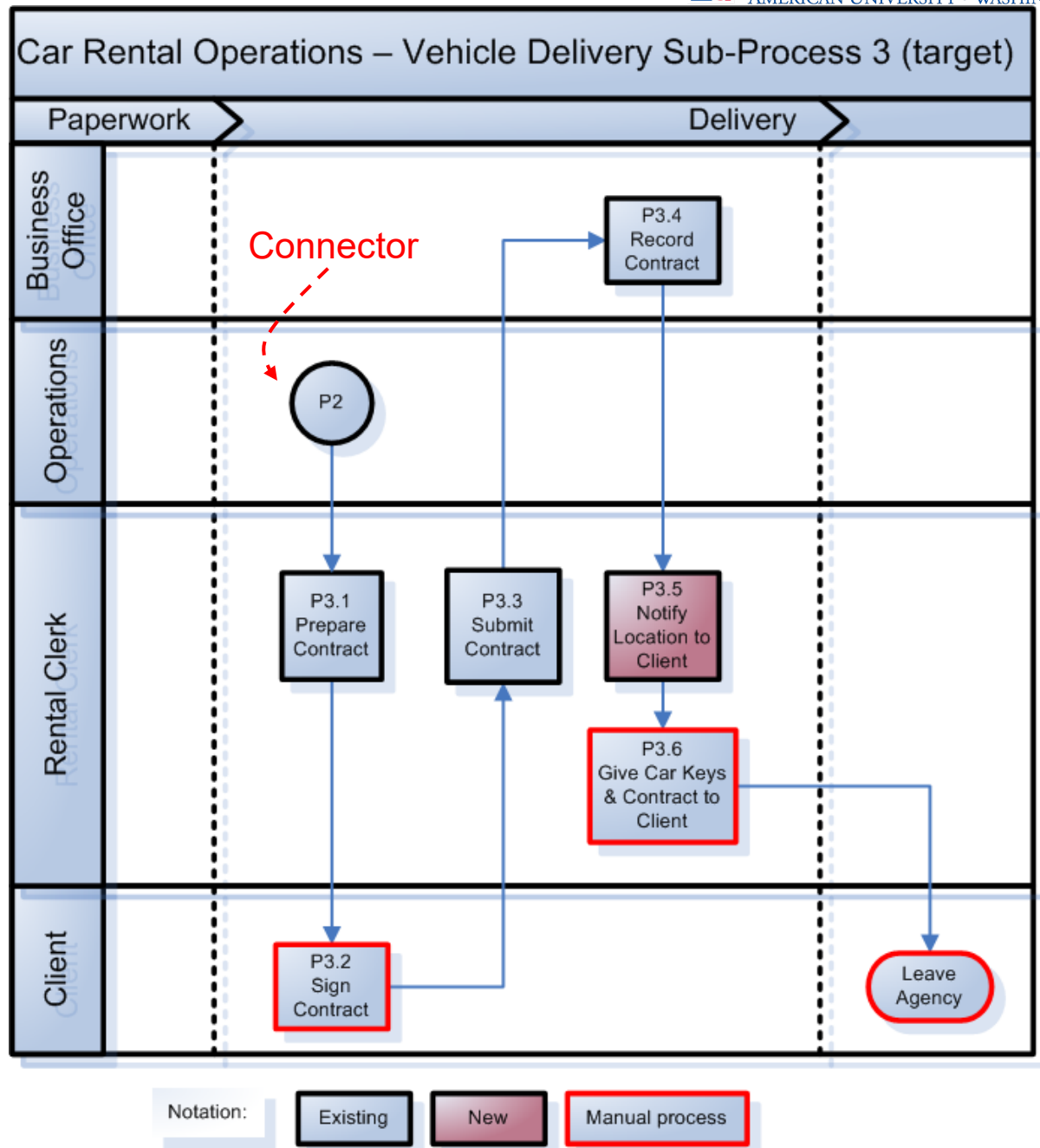
BPM Example – Proposed



BPM Example – High Level Diagram



BPM Example: Defined Process Details



Tools

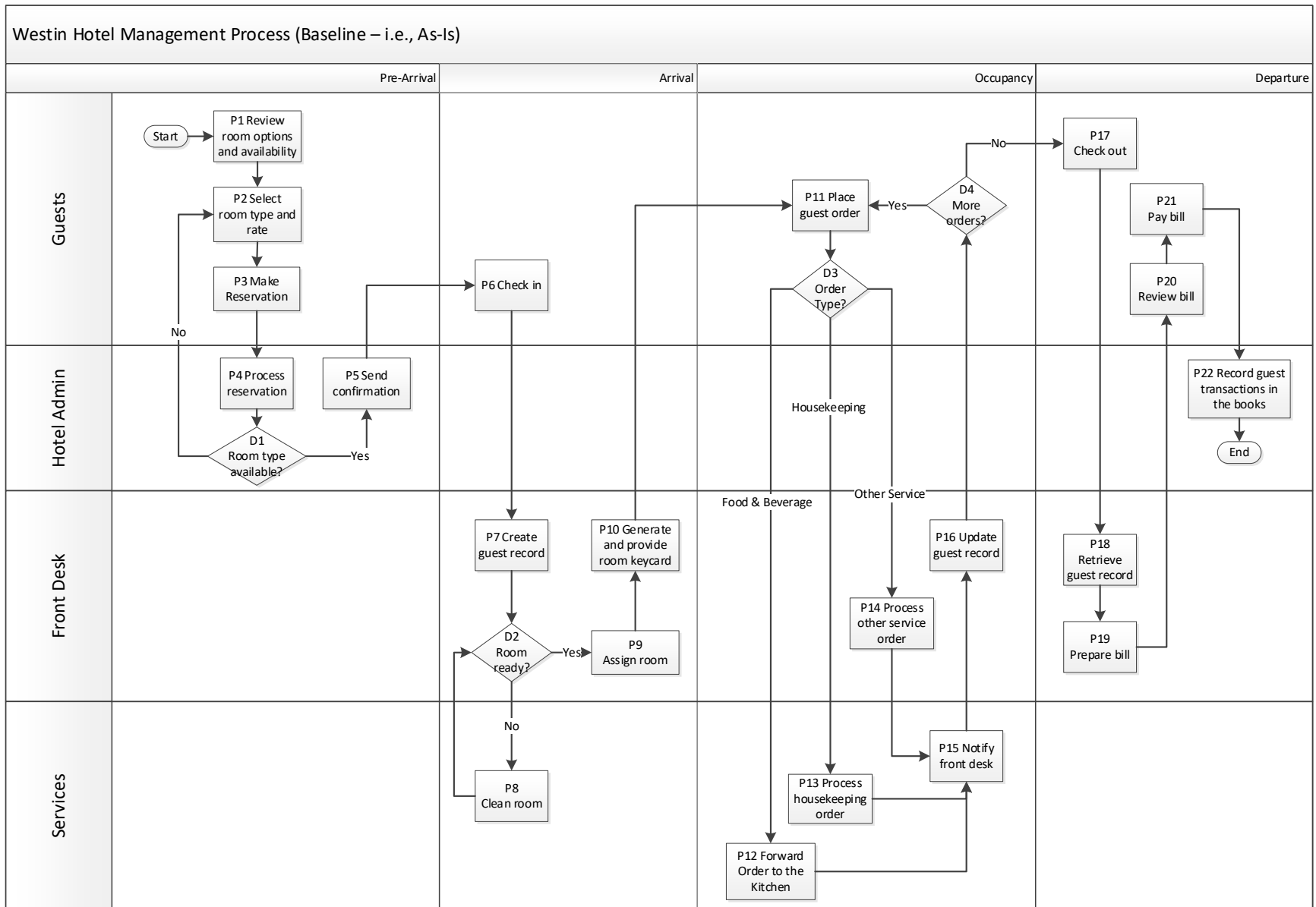
- diagrams.net (<https://app.diagrams.net/>)
 - Free, simple and intuitive business diagramming
 - Not just for BPMs but many other diagrams
 - Online, no need to create an account
- Lucidchart (<https://lucid.co/lucidchart>)
 - Free (with substantial limitations) and paid versions
 - Need to create an account
- Microsoft Visio (<https://www.microsoft.com/en-us/microsoft-365/visio>)
 - Excellent diagramming tool, paid w/free trial versions
- Smart Draw (<https://www.smartdraw.com/>)
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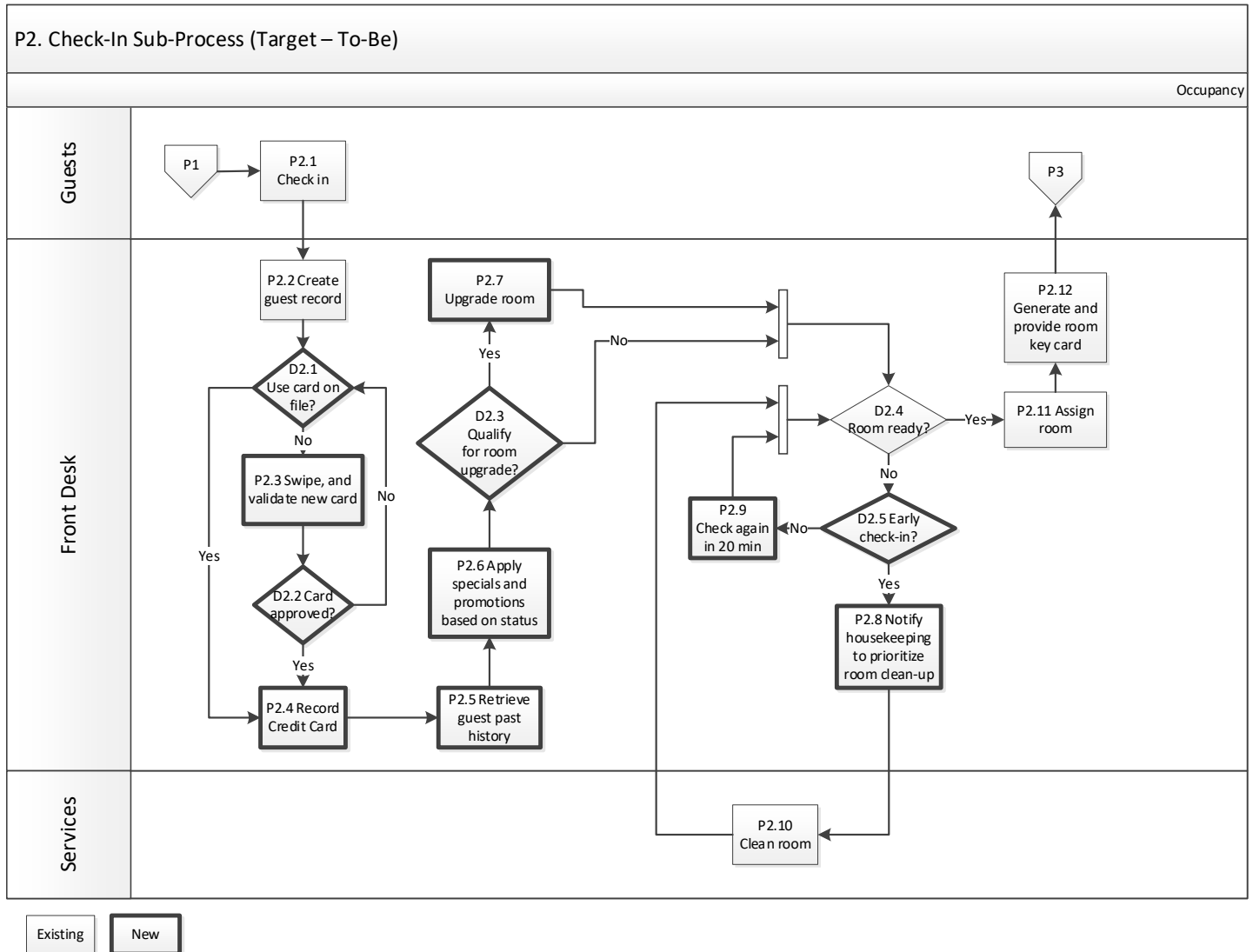
BPM Examples



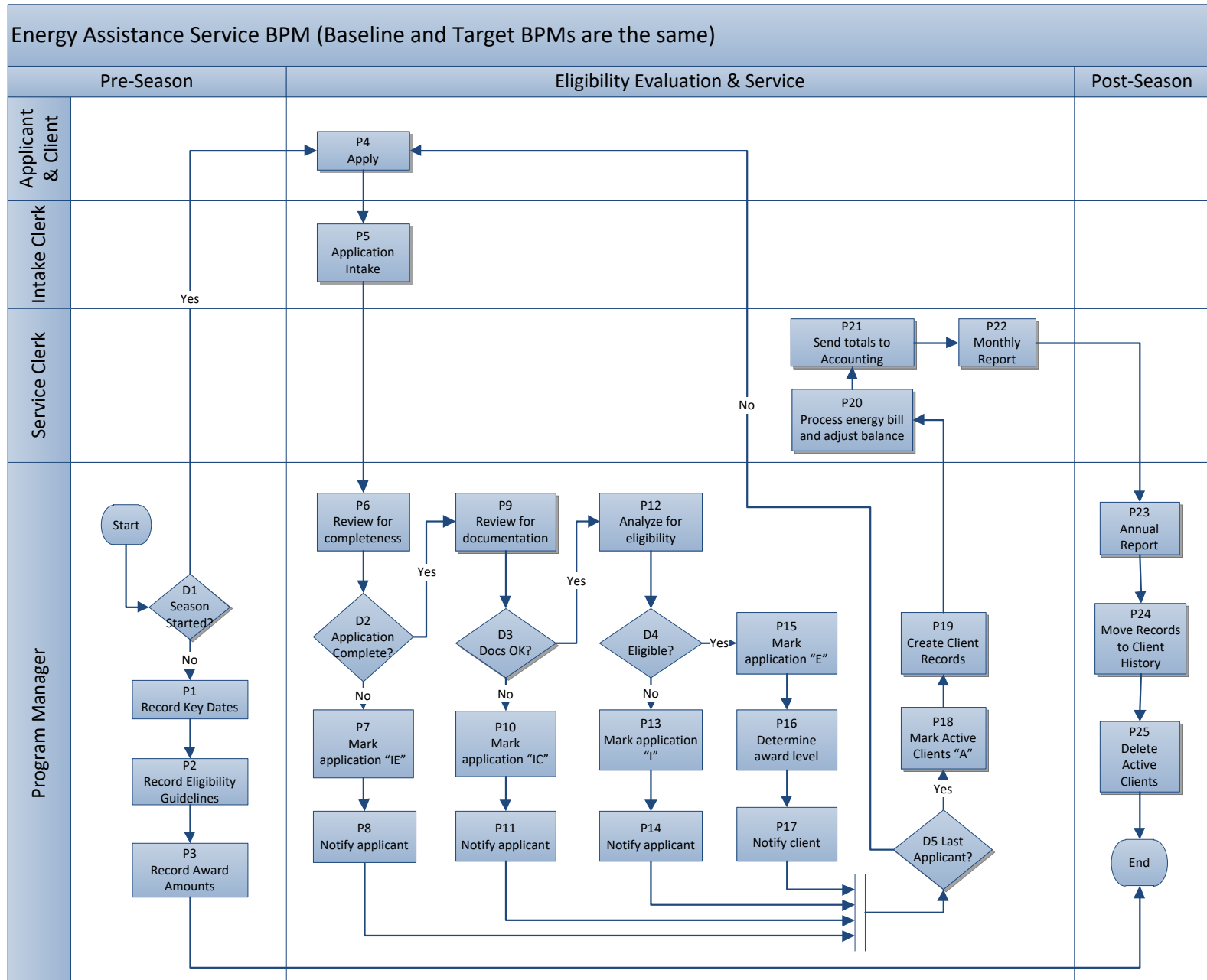
BPM Example: Westin Lima Hotel Management



BPM Example: Check-In Process Improvement



BPM Example: Energy Assistance Process





Questions ?